DATA SCIENCE FOR MARKETING

Lisbon Tourism

Prof: Nuno António

Group 10: Beatriz Santos (20221937), Bernardo Teixeira (20250831), Manuel Freudenthal (20250835) & Maria Inês Castro (20221842)

1. EXECUTIVE SUMMARY

Turismo de Lisboa aims to better understand visitor experience and use online feedback to inform marketing and operational decisions. This project addresses that goal by transforming Tripadvisor reviews of Lisbon's attractions into structured insights that help identify experience drivers, emerging issues, and opportunities for targeted action.

Following the CRISP-DM framework, the analysis begins with a careful assessment of the available data and its limitations, followed by data preparation to ensure consistency and usability. The project then combines exploratory analysis with review analytics to examine how visitor satisfaction varies across attractions, time periods, and experience dimensions. Textual content is leveraged to move beyond star ratings, allowing the identification of key themes, patterns of visitor behavior, and signals of dissatisfaction that may not be immediately visible in aggregate scores.

The evaluation phase validates the relevance and robustness of these insights, ensuring they are meaningful for decision-making and aligned with Turismo de Lisboa's strategic objectives. Overall, the project delivers a practical, review-driven framework that enables more proactive monitoring of visitor experience, supports prioritization of high-impact interventions, and contributes to more informed and sustainable tourism management.

2. BACKGROUND

Tourism plays a central role in Lisbon's economic, cultural, and social development. As one of Europe's most visited city destinations, Lisbon attracts millions of visitors each year to its historical monuments, museums, viewpoints, tours, and cultural experiences.

In this context, maintaining a high-quality visitor experience is essential not only to sustain demand, but also to protect the city's reputation, manage overcrowding, and ensure that tourism growth remains sustainable for residents and visitors alike. In recent years, online travel platforms, particularly Tripadvisor, have become a primary source of information for tourists when planning visits and selecting attractions. Visitor reviews published on these platforms strongly influence expectations, choices, and perceptions of value. At the same time, they provide a rich and largely untapped source of feedback for destination management organizations.

Turismo de Lisboa (LTA) seeks to move beyond simple star ratings and use this large volume of user-generated content as a strategic decision-support tool. Rather than focusing only on promotional performance, LTA aims to understand why visitors are satisfied or dissatisfied, where operational or service issues emerge, and which interventions would have the greatest impact on visitor experience in upcoming tourist seasons.

This project positions our team as LTA's analytics partner, applying data science techniques within the CRISP-DM framework to transform Tripadvisor reviews into actionable marketing and operational insights. The analysis focuses on diagnosing experience quality at both the city and attraction level, identifying the key drivers of satisfaction and dissatisfaction, and supporting evidence-based prioritization of improvement initiatives.

The analysis is based on three datasets provided for the project. The main dataset contains Tripadvisor reviews of Lisbon's attractions, including star ratings, review text, visit dates, reviewer characteristics, and information on management responses. This is complemented by an attraction metadata dataset with descriptive attributes such as category, location, global ratings, and review volume, as well as a curated list of attractions defining the scope of the study. Together, these datasets enable a comprehensive view of visitor experience across attractions, time periods, and visitor segments.

3. PROBLEM

While Lisbon's attractions generate a large volume of TripAdvisor reviews, Turismo de Lisboa currently lacks a systematic, data-driven approach to turn this feedback into actionable guidance for marketing and operational decisions. Relying primarily on aggregate ratings masks important differences across attractions, seasons, visitor segments, and the underlying drivers of satisfaction and dissatisfaction.

This project addresses the challenge of transforming large-scale review data into decision-ready insights that support marketing prioritization, experience improvement, and strategic planning. By combining structured information with review text, and by accounting for biases and seasonality, the analysis aims to identify key experience drivers and prioritize high-impact interventions that strengthen Lisbon's visitor experience and competitive positioning.

4. METHODS AND RESULTS

4.1. Business Understanding

Lisbon Tourism Authority (LTA) is responsible for promoting Lisbon as a high-quality, competitive, and sustainable tourist destination. Online platforms such as Tripadvisor play a critical role in shaping visitors' expectations and decisions, as well as in reflecting real experiences after the visit. Thousands of user-generated reviews capture perceptions about attractions, services, value for money, crowding, accessibility, and overall satisfaction.

However, these reviews are largely unstructured and difficult to translate into concrete managerial and marketing actions. Our role is to aid LTA in transforming dispersed review data into actionable insights that support better decision-making in marketing, operations, and visitor experience management, especially ahead of upcoming tourism seasons marked by strong demand and capacity constraints.

From a business perspective, the goal of **sustained tourism development** requires that growth in visitor numbers does not come at the expense of experience quality or destination reputation. Measuring and tracking satisfaction over time, through KPI like Overall Satisfaction Score, and experience-related indicators like Experience Quality Index, allows LTA to assess whether tourism development is balanced and sustainable. Similarly, the **promotion of Lisbon as a leisure destination** relies on identifying the attractions and non-business experiences that generate the highest levels of satisfaction and emotional engagement, through indicators like Topic Sentiment Score per visit type, so that these strengths can be amplified through marketing content and communication strategies.

At the same time, **Lisbon's positioning as a destination for congresses, fairs, and incentive travel** depends on the delivery of reliable, well-organized, and accessible experiences. Visitor feedback related to logistics, staff quality, accessibility, crowding, and clarity of information, through indicators like Accessibility Score or Crowding Pressure Index, provides valuable signals about Lisbon's readiness to meet the expectations of business-oriented visitors. Finally, LTA's role in **providing information and support to tourists** is directly reflected in reviews mentioning signage, transport guidance, staff helpfulness, and overall ease of visiting attractions, which translate into KPIs like Staff Efficiency Index.

By translating visitor reviews into structured indicators and insights, the project aims to support evidence-based decision-making, prioritize high-impact actions for the next tourism season, and provide clear, data-driven recommendations that enhance visitor satisfaction while reinforcing Lisbon's long-term attractiveness and sustainability as a tourism destination.

4.2. Data Understanding

Before extracting insights or making recommendations, it is essential to understand what the data represents, what it does not, and how visitor opinions are expressed on TripAdvisor. This section assesses the structure, coverage, and behavioral patterns in the review data to ensure that later findings are interpreted correctly and responsibly.

4.2.1. Data Organization

Before extracting insights or making recommendations, it is essential to understand what the data represents, what it does not, and how visitor opinions are expressed on TripAdvisor. This section assesses the structure, coverage, and behavioral patterns in the review data to ensure that later findings are interpreted correctly and responsibly.

For that, we decided to group our variables into thematic categories based on what they represent. This organization helps clarify the role of each variable in describing the visitor experience and guides more focused and segmented analyses in later stages.

- **Review & Experience Variables**

Across Lisbon's attractions, visitor satisfaction is exceptionally high. Ratings are overwhelmingly concentrated at the top of the scale, with the large majority of visitors awarding 5-star reviews. Low ratings exist but are comparatively rare, and overall variability is limited.

This pattern suggests that Lisbon enjoys a strong baseline reputation among visitors.

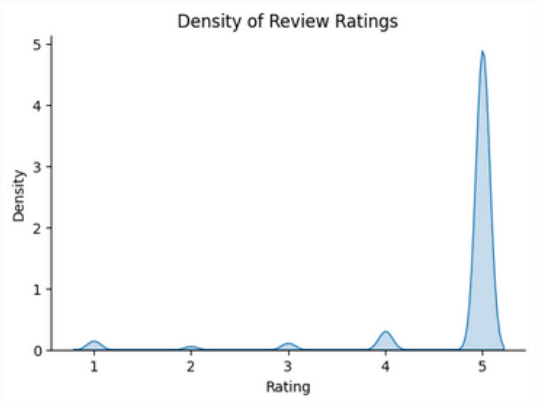

Figure 1. Distribution of TripAdvisor ratings

However, it also implies that average ratings alone are insufficient to identify improvement opportunities, as they naturally cluster near the maximum score.

A clear behavioral pattern emerges when comparing review length with satisfaction. Lower-rated reviews tend to be substantially longer, while highly positive reviews are typically brief. This indicates that dissatisfied visitors are more motivated to explain, justify, and detail their experiences, often highlighting specific operational issues.

Responses are not distributed evenly: attractions tend to reply more frequently to extreme reviews, particularly very positive ones and, to a lesser extent, very negative ones. Taking this into consideration, we analyzed that review management is currently used more as a reputation reinforcement tool than as a systematic service-recovery mechanism. While positive engagement strengthens brand image, unanswered negative reviews may represent missed opportunities to address dissatisfaction publicly.

- **Visitor Profile Variables**

Most reviews come from leisure visitors traveling as couples, families, or friends, which together account for nearly 90% of all visits. Despite differences in visit context, satisfaction levels are remarkably consistent across visit types, with only minimal variation in average ratings. This indicates that Lisbon's attractions deliver a broadly similar experience regardless of group composition.

Given this homogeneity, visit type is more useful for segmentation and tailored communication than for identifying dissatisfaction drivers.

Geographic information is highly incomplete and inconsistent, with over half of reviewers lacking a usable country identifier. Additionally, location data often reflects cities or regions rather than standardized country names, limiting its reliability.

As a result, geographic segmentation based on this variable would risk misleading conclusions unless extensive cleaning and normalization are performed. This limitation is important to acknowledge, as it constrains the precision of market-level insights.

- **Temporal Variables**

Review volume shows clear seasonality, increasing from spring through summer and declining during winter months. This confirms that visitor feedback is not evenly distributed across the year, with peak-season experiences more heavily represented in the data.

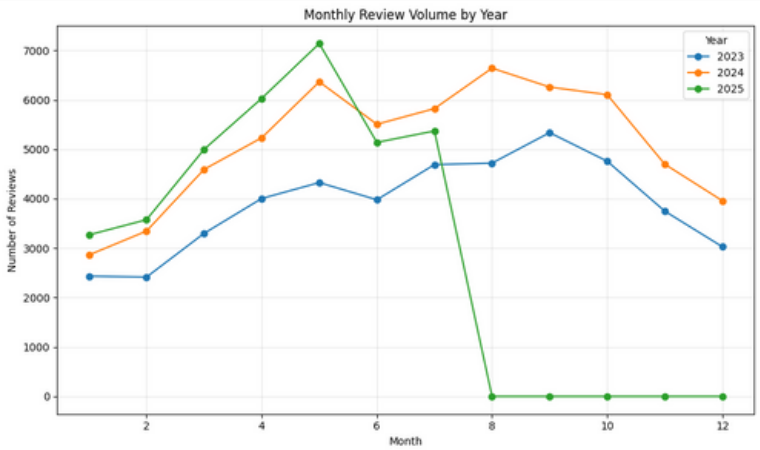

Figure 2. Monthly review volume by year

Despite strong seasonal fluctuations in review volume, average satisfaction remains remarkably stable throughout the year. This suggests that, at an aggregate level, visitor experience quality is resilient to seasonal pressures such as crowding or weather. However, given the strong skew toward high ratings, this stability should not be interpreted as an absence of seasonal issues, but rather as a limitation of star ratings in capturing operational stress points.

- **Attraction & Supply-Side Variables**

Visitor attention is highly concentrated among a small number of attractions. A handful of sites account for a disproportionately large share of all reviews, with one attraction alone representing nearly a quarter of total review volume.
This uneven distribution implies that:

- A few attractions disproportionately shape Lisbon's online reputation;
- Many smaller attractions suffer from low visibility rather than poor performance;
- City-level averages may be driven by a limited subset of experiences.

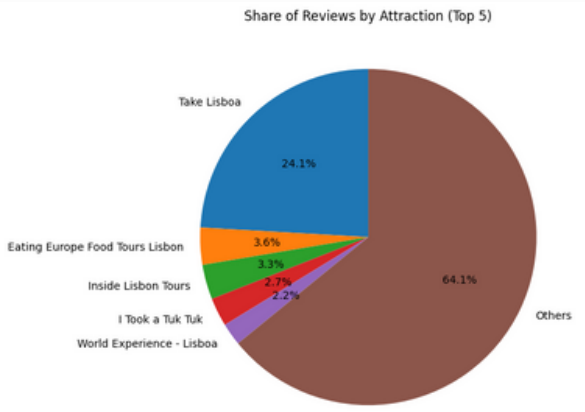

Figure 3. Share of reviews by attraction

Attraction-level global ratings broadly align with individual review scores, indicating that they are a reasonable summary of overall satisfaction.

- **Engagement & Interaction Variables**

Both reviewer activity (contributions) and review engagement (likes) are highly concentrated among a small minority of users and reviews. Most reviews receive little to no interaction, while a few generate disproportionate attention. Interestingly, moderate and critical reviews tend to receive more likes than purely positive ones, suggesting that balanced or negative feedback is perceived as more helpful by other users. Leading us to conclude that influence on perception is not driven by enthusiasm alone, but by perceived credibility and informational value.

4.3 Exploratory Data Analysis

The purpose of this exploratory analysis is to assess data quality, consistency, and behavioral patterns before moving into deeper analytical modeling. Rather than identifying statistical relationships, this section ensures that subsequent insights are credible, interpretable, and aligned with real-world visitor behavior.
To support this goal, variables were grouped into numerical, categorical, temporal, text-based, and identifier subsets, allowing each type to be assessed using appropriate lenses.

4.3.1. Missing Values

Several variables contain missing values, but not all missingness represents a data problem.
In particular, the absence of management replies (replyText, replyDate) simply reflects that many reviews did not receive a response, which is a meaningful behavioral signal rather than an error.
Other variables present more substantial gaps:

- Country, visitMonth, and visitType show a high proportion of missing values;
- Geographic information is especially incomplete and inconsistently formatted, limiting its reliability for segmentation without extensive cleaning;
- Visit context variables (such as visit type and visit month) remain useful but must be interpreted cautiously due to partial coverage.

4.3.2. Outliers and Distribution Shape

Across the numerical variables, extreme values are common but mostly reflect how visitors actually behave on TripAdvisor rather than data quality issues. Ratings are naturally limited to a 1–5 scale, so very high or low values are expected. Engagement indicators such as likes and contributions show that most users interact very little, while a small group is much more active. Review length also varies widely, with longer comments often associated with more detailed or critical experiences.
At the attraction level, review volumes and rankings are unevenly distributed, as a small number of highly popular attractions receive most of the attention. Overall, these patterns reflect normal platform dynamics and should be seen as informative characteristics of visitor behavior, not anomalies to be removed.

4.3.3. Inconsistencies and Variable Coherence

Several variables were reviewed to ensure they were coherent and meaningful for analysis. The country variable shows a high level of inconsistency, as it often reflects cities, regions, or states instead of standardized country names, which limits its immediate usefulness for geographic segmentation. In contrast, the visit type variable is clearly defined and consistent across observations. The variables num_reviews and reviewCount refer to the same underlying concept and differ only marginally, indicating redundancy rather than conflicting information. Engagement indicators such as likes and contributions are generally very low, reflecting typical user behavior on the platform rather than data quality problems. Finally, most reviews are written in the same month as the visit, although a small number of records show minor timing inconsistencies. Overall, these issues are limited and well understood, making them straightforward to handle in the data preparation stage.

4.3.4 Duplicates

Only a very small number of duplicate reviews were identified. These correspond to the same reviewer posting effectively identical feedback for the same attraction, introducing redundancy but not distortion.

Additionally, repeated management replies were observed. These are not errors, but standardized responses commonly used for positive reviews, reflecting efficiency rather than negligence.
Given their limited number and clear interpretation, duplicates can be safely removed in the data preparation phase without affecting the integrity of the analysis.

4.4. Data Cleaning

The data cleaning phase focused on resolving the main quality issues identified during data understanding, with the objective of improving consistency, reliability, and usability for subsequent analysis.

Duplicate reviews were removed to eliminate redundant information, using a combination of a tripadvisor_id, reviewer, and review text. Redundant variables capturing the same concept were simplified by retaining the most complete and reliable version. Missing values were handled selectively: variables with negligible missingness were imputed using simple, robust strategies, while variables with high incompleteness or low analytical relevance were excluded to avoid introducing bias.

Outliers were assessed but not removed, as extreme values largely reflect genuine visitor behavior, engagement patterns, and attraction popularity rather than data errors. Instead, these variables were retained for later transformation during feature engineering. Finally, minor incoherences were corrected, including misleading variable names and a small number of temporally inconsistent records where reviews preceded the recorded visit date.

Overall, the cleaning process preserved the richness of the data while ensuring structural coherence and analytical readiness, providing a solid foundation for the modeling and insight generation phases.

4.5. Data Tranformation

The transformations phase aims to reduce noise, address skewness, and translate raw platform metrics into analytically meaningful indicators.

4.5.1. Feature Engineering

The transformations phase aims to reduce noise, address skewness, and translate raw platform metrics into analytically meaningful indicators.

- **Multidimensional Features: Attraction Ranking**

Attraction ranking information is inherently relative: a ranking position only has meaning when interpreted in the context of how many attractions exist within the same category. For example, being ranked 10th has a very different implication in a category with 20 attractions than in one with 200. To address this, ranking position and category size were combined into a single relative ranking indicator. This transformation expresses how well an attraction performs within its own category on a standardized scale, enabling meaningful comparison across categories of different sizes.

- **Binarization of Engagement: Review Likes**

Rather than focusing on the exact number of likes, which adds limited interpretive value in such a skewed context, we decided to transform this variable into a simple binary indicator reflecting whether a review received any likes at all. This approach emphasizes the presence of engagement rather than its intensity, making the signal easier to interpret and more aligned with the objective of identifying reviews that resonate with other users.

- **Seasonal Features: Visit Timing**

Tourism activity in Lisbon varies substantially throughout the year, influencing both visitor behavior and operational conditions at attractions. To capture these dynamics in a clear and intuitive way, visit dates were grouped into three seasonal periods: Peak, Shoulder, and Off-Peak. This transformation translates calendar information into meaningful demand contexts, allowing comparisons of visitor experience, engagement, and management behavior across different levels of tourism intensity. By doing so, seasonality becomes an explicit analytical dimension rather than a background effect.

- **Management Responsiveness: Fast Replies**

To capture how actively attractions manage their online reputation, a new variable was created to reflect response speed to visitor reviews. This threshold provides a simple and intuitive distinction between timely and delayed responses, allowing responsiveness to be interpreted as a behavioral signal rather than a precise time measurement.

- **Text Mining and Experience Signals**

Review text was transformed into structured variables that capture underlying experience signals. Therefore, entiment analysis was applied to extract an overall emotional tone for each review. While sentiment scores closely follow rating levels, they also add nuance by differentiating reviews within the same star category, particularly among highly rated experiences.

The analysis shows that operational factors, especially staff interactions and waiting time, are disproportionately present in negative reviews and are more likely to elicit management responses.

Overall, the data transformation process translated raw Tripadvisor data into analytically meaningful constructs that reflect how visitors engage, how attractions perform, and how experiences are managed over time. By moving beyond raw counts and ratings, the engineered features capture differences in engagement intensity, seasonal demand, reputational exposure, and management responsiveness. This enriched and structured dataset provides a solid foundation for modeling visitor behavior and supports the generation of actionable insights in the subsequent analysis phases.

4.6. RFM Analysis

4.6.1. RFM Model Adaptation

The traditional RFM (Recency, Frequency, Monetary) framework is widely used to segment customers based on transactional value. However, because Tripadvisor data does not include direct spending information, the Monetary dimension was redefined to reflect reputational and engagement value rather than financial value. This adaptation allows the RFM model to remain conceptually consistent while being meaningful in a review-based context.

- **Reputational Impact: Impact Rating**

Not all reviews influence an attraction's reputation equally. Lower-rated reviews (three stars or below) tend to attract more attention from potential visitors and often prompt management responses. To capture this asymmetry, we created an impact rating variable to flag reviews with higher potential reputational consequences. This feature highlights reviews that are more likely to influence perceptions and decision-making among future visitors.

- **Engagement-Based Monetary Value**

The adapted Monetary component represents the value generated by a review in terms of visibility, engagement, and managerial attention. It combines four complementary signals:

  - whether the review received likes (peer engagement);
  - whether it received a management reply (reputational relevance);
  - the overall popularity of the attraction (baseline visibility);
  - and whether the review had high reputational impact (low rating).

This composite score allows reviews to be differentiated not only by sentiment, but by their broader influence on the platform. Higher values therefore represent reviews that contribute more strongly to an attraction's online presence and reputation.

- **RFM Dimensions Interpretation**
  - Recency captures how recently a reviewer visited an attraction, reflecting current relevance and likelihood of future engagement;
  - Frequency measures how often a reviewer interacted with the same attraction, indicating loyalty or repeated interest;
  - Monetary (Engagement Value) reflects the cumulative reputational and engagement impact generated by the reviewer's activity.

Together, these dimensions provide a multidimensional view of reviewer value that goes beyond simple activity counts.

4.6.1. Distribution Analysis of RFM Metrics

To evaluate the behavior captured by the RFM variables, we first examine their empirical distributions through histograms of Recency, Frequency, and Monetary values. This step is essential to assess whether the variables exhibit sufficient variability to meaningfully differentiate reviewer profiles and to understand the overall structure of reviewer engagement.

- **Recency**

The recency distribution is strongly right-skewed. Most reviewers have relatively low recency values, indicating that they visited attractions fairly recently. However, a long tail of higher values shows that a non-negligible share of reviewers has not interacted for a long time. This spread confirms that recency is effective in distinguishing active reviewers from inactive or disengaged ones.

- **Frequency**

Frequency is heavily concentrated at very low values. The majority of reviewers visited an attraction only once, while a small number of reviewers appear multiple times. These highly active reviewers form a thin but important tail of the distribution and represent repeated engagement rather than one-off visits, which is typical in tourism platforms.

- **Monetary**

The monetary dimension is also highly right-skewed. Most reviewers generate limited engagement, while a small minority produces very high reputational impact through likes, replies, and visibility. This indicates that engagement on the platform is unevenly distributed, with a few reviewers contributing disproportionately to attraction reputation.

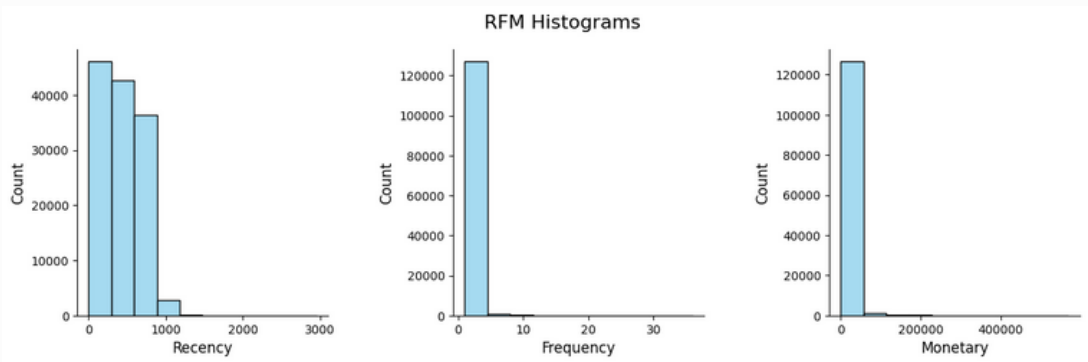

Figure 4. RFM Histograms Distribution

4.6.2. RFM Segment Profiling of Reviewer Behavior

For this stage of the RFM analysis, we adopted a segmentation-oriented perspective aimed at translating RFM scores into interpretable reviewer profiles. This approach provides an initial, high-level segmentation that helps distinguish different types of reviewer behavior and their relative value for attractions on Tripadvisor.

Type of reviewer	Behavioral Profile	Key Takeaway
High-value and active reviewers	Recent visits, repeat activity, high engagement	Small but critical group driving visibility and reputation
Recent but low-engagement	Recent visit, single interaction, low impact	Largest opportunity to increase engagement and influence
Loyal but inactive	Past repeat visits, no recent activity	Risk of disengagement; potential reactivation target
Low-value	Old visits, single interaction, minimal engagement	Limited strategic value; low priority for interventions

- **High-value and active reviewers**: A very small group, but they generate a disproportionate share of engagement and visibility, making them the most strategically important reviewers.
- **Recent but low-engagement reviewers**: The largest active segment; they visit recently but have limited impact, representing the main opportunity to increase engagement and influence.
- **Loyal but inactive reviewers**: Previously engaged reviewers who are no longer active, signaling potential churn and re-engagement opportunities.
- **Low-value reviewers**: A sizable group with minimal activity and impact, contributing little to visibility and offering limited strategic value.

4.6.3. 3D Scatter Plot Analysis

The 3D RFM plot further clarifies how reviewer behavior clusters across the three dimensions simultaneously. While many reviewers are grouped near low Frequency and Monetary values, a distinct set of high-value reviewers emerges, combining recent activity with high engagement and strong reputational impact. This visualization helps distinguish between reviewers who are simply recent and those who are genuinely influential.

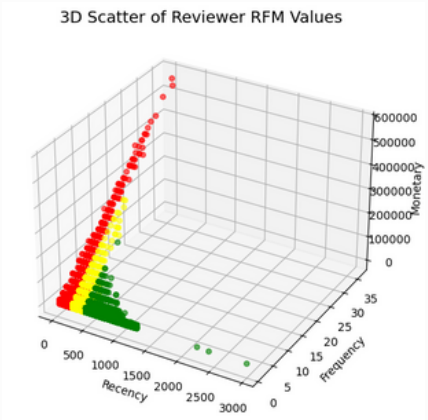

Figure 5. 3D Scatter of Reviewers

4.6.4. Treemap: Reviewers by RFM Segment

To complement the previous analyses, a treemap was used to show how reviewers are distributed across RFM segments. Each rectangle represents an RFM segment, and its size reflects the number of unique reviewers in that segment. This allows for an immediate comparison of segment importance based on volume rather than averages.

Key insights from the treemap:

- Most reviewers fall into low-engagement segments, indicating that platform activity is dominated by occasional or low-impact contributors.
- High-value segments occupy much smaller areas, confirming that highly engaged and influential reviewers are rare.
- Despite their small size, high RFM segments are strategically important because they generate disproportionate visibility and engagement.
- The clear imbalance across segments highlights strong behavioral differences and supports the relevance of RFM segmentation for understanding reviewer value.

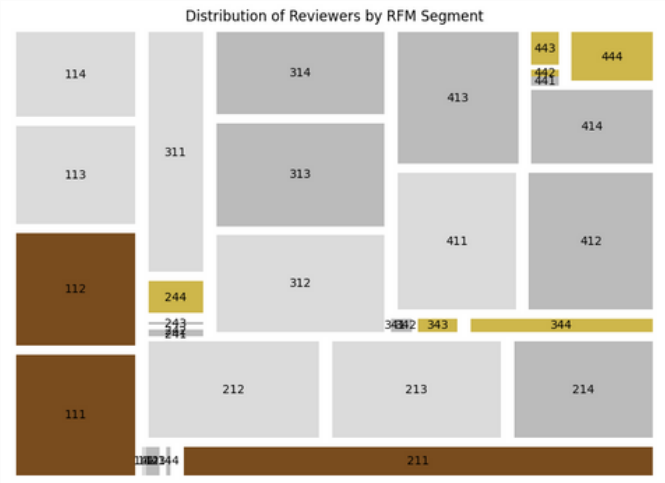

Figure 6. Treemap of Reviewers

4.7. Assessing Segmentation Quality

To evaluate the quality of the RFM segmentation, we performed a simple temporal validation to check whether reviewers classified as higher value based on past behavior also generate higher value in the future. The data was split into a training period (to compute RFM scores) and a test period (to observe future monetary impact).

The results show a clear upward trend: as RFM scores increase, average future value also increases. Low-RFM segments generate little future engagement, while high-RFM segments, especially those with strong Frequency and Monetary scores, produce substantially higher future impact.

This confirms that the RFM segmentation is not only descriptive but also predictive, providing a reliable basis for understanding reviewer value and supporting data-driven decisions.

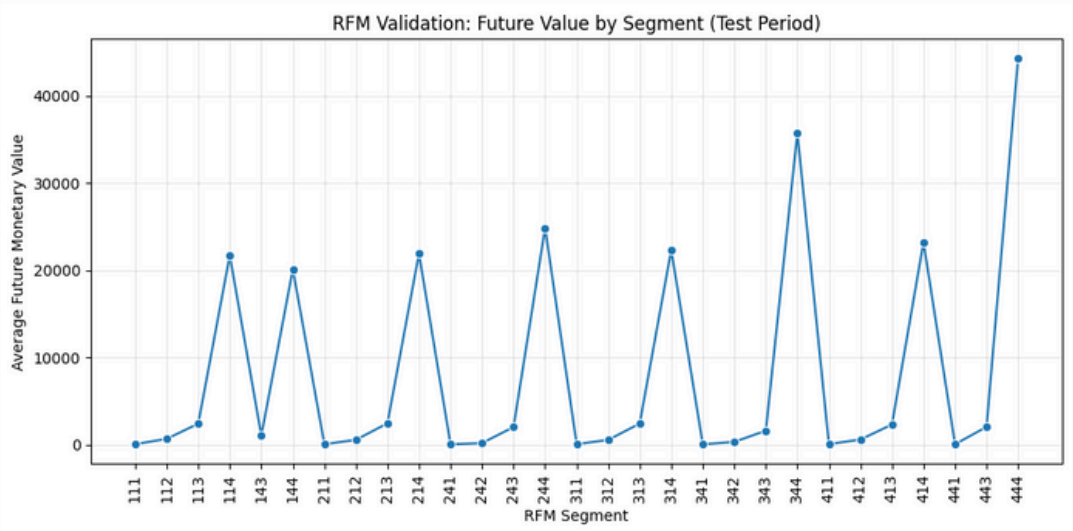

Figure 7. RFM Validation: Future Value by Segment

4.6.6. RFM Interpretation and Campaign Strategy

To make the RFM segmentation actionable, we translated the recency dimension into clear campaign groups. The goal is to move from descriptive segmentation to practical decision-making by identifying which reviewers should be maintained, retained, or re-activated. Based on how recently a reviewer interacted, reviewers were classified into three intuitive campaign types:

- **Active:** recently engaged reviewers
- **At Risk:** reviewers whose activity is declining
- **Churned:** reviewers who have not interacted for a long time

- Most reviewers fall into the "**Churned"** group, indicating widespread disengagement;
- "**Active"** reviewers represent a smaller but valuable group with strong potential for continued engagement;
- "**At Risk"** reviewers are the smallest segment, making them a clear target for timely retention actions.

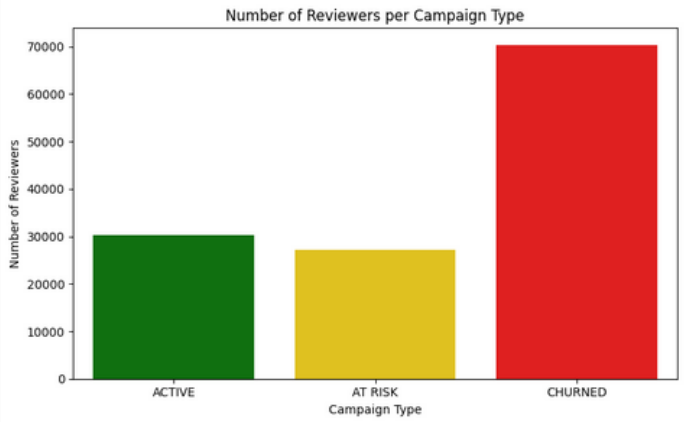

Figure 8. Distribution of reviewers per Campaign Type

Our team concluded about the urgency of applying these campaigns:

- **Retention** and **reactivation** should be prioritized over broad promotional efforts;
- Active reviewers are best suited for reinforcement and loyalty-focused actions;
- At risk reviewers offer the highest short-term return from targeted retention campaigns;
- Churned reviewers require selective reactivation strategies rather than mass outreach.

4.7. Similarity Measures

In this section, similarity measures are used to uncover patterns of resemblance between attractions based on how they are experienced, managed, and engaged with on Tripadvisor. Rather than focusing on individual reviews, the objective is to compare attractions holistically, capturing similarities in visitor satisfaction, visibility, engagement intensity, and management behavior.

This approach allows us to address two practical questions:

- Which attractions exhibit similar experiential and engagement profiles?
- Which attractions could be recommended to visitors based on the ones they have already visited?

Our unit of analysis is the reviewer–attraction interaction, representing how a specific reviewer evaluated a specific attraction. This structure preserves individual preferences while allowing aggregation to higher-level patterns.

4.7.1. Attraction-Level Feature Aggregation

To meaningfully compare attractions, individual reviews were consolidated into a set of attraction-level indicators that capture how each attraction is experienced, perceived, and managed on Tripadvisor. Rather than relying solely on average ratings, we combined multiple dimensions that together describe an attraction's overall profile: perceived quality, relative standing within its category, visibility on the platform, visitor engagement, responsiveness to feedback, and potential reputational impact.

This aggregation allows us to move beyond a narrow view of performance. Two attractions may have similar ratings but differ substantially in how visible they are, how visitors interact with them, or how actively they manage their online reputation. By integrating these dimensions, similarity reflects the full experiential and operational footprint of an attraction, not just how well it scores.

4.7.2. Similarity Computation

Similarity between attractions was computed to reflect overall behavioral and experiential patterns rather than popularity alone. Features were first made comparable so that no single variable dominated the comparison. Two attractions are considered similar when they display comparable profiles in terms of quality, engagement, and management behavior, even if their visibility or review volume differs. This allows meaningful similarities to be identified beyond raw scale.

4.7.3. Recommendation Logic

Similarity scores were then used to demonstrate a basic recommendation approach. Starting from an attraction already visited by a reviewer, similar attractions were identified and those not yet visited were recommended. This follows a simple logic: visitors are more likely to enjoy attractions that resemble those they have already experienced, prioritizing relevance over popularity.

4.7.3. Interpretation of Results

The recommendations consistently identify attractions with similar experience quality, engagement patterns, and management behavior. Importantly, some suggested attractions have high experiential similarity despite lower visibility or fewer reviews, indicating that similarity is driven by underlying experience and behavior rather than popularity alone.

This allows the analysis to reveal meaningful connections between attractions, supporting the discoverability of less-known sites, the grouping of attractions into coherent experiential clusters, and recommendation strategies that promote diversity instead of reinforcing existing popularity biases.

Overall, similarity analysis complements the RFM segmentation by shifting the focus from individual reviewers to relationships between attractions, enabling both strategic benchmarking and visitor-centric recommendations aligned with experience quality and destination management objectives.

4.8. Market Basket Analysis

Market basket analysis was used to identify which review characteristics tend to appear together. Instead of products in a shopping cart, each "basket" represents a single review and contains attributes such as engagement signals, management responses, experiential mentions, seasonality, and attraction type. This approach helps uncover recurring patterns in how visitors describe and interact with attractions.

4.8.1. Association Rule Mining

Reviews were transformed into a binary format indicating the presence or absence of specific attributes (e.g., likes, replies, mentions of staff or price, season, attraction category). This allowed us to examine which features frequently co-occur within the same review.

The most frequent attributes in the dataset relate to:

- Attraction category (especially tours and activities);
- Mentions of staff;
- Seasonality.

This suggests that visitor feedback is strongly shaped by the type of attraction, interpersonal service quality, and timing of the visit.

More informative patterns emerge when attributes are considered jointly. For example, reviews that mention staff are more likely to receive management replies, and fast replies often occur in reviews that already triggered engagement or operational feedback. These combinations highlight how certain types of visitor comments are more likely to prompt managerial action.

4.8.2. Interpretation of Association Rules

Overall, most associations are relatively weak, indicating that many review attributes occur together simply because they are common, not because one strongly drives another. This is particularly true for seasonality, which appears frequently but does not substantially change the likelihood of other attributes being present.

However, several meaningful patterns stand out:

- Staff-related mentions frequently co-occur with management replies, suggesting that operational feedback is more likely to trigger responses;
- Fast replies tend to appear in reviews that already received attention (likes or replies), pointing to targeted rather than uniform responsiveness;
- Attraction category plays a stronger role than seasonality in shaping review content, especially for tours and activities.

These findings indicate that management engagement is not random: it is more closely linked to the content and perceived importance of reviews than to external factors such as time of year.

4.8.3. Implications for Destination Management

Market basket analysis complements similarity and segmentation analyses by focusing on co-occurring review signals rather than averages. It helps identify:

- Which types of reviews are most likely to receive a management response;
- Which visitor mentions or experience themes tend to appear together in the same reviews;
- Whether management replies are driven by specific issues rather than applied consistently across reviews.

From a destination-management perspective, this provides guidance on:

- What kinds of feedback trigger replies;
- Which problems visitors often mention together;
- Whether replies are strategic or mostly reactive.

Overall, market basket analysis reveals how engagement, experience signals, and management behavior interact within individual reviews, offering actionable insight into the dynamics of visitor feedback on the platform.

5. EVALUATION

In this evaluation phase, we focused on examining whether the analytical results obtained in earlier stages are statistically valid, robust, and aligned with the project objectives. This section focuses on hypothesis testing and interpretability to ensure that insights from text mining, RFM analysis, similarity measures, and market basket analysis are both statistically valuable and useful for decision-making.

5.1. Hypothesis Testing

H1 - Sentiment as an Early-Warning Signal: Do changes in review sentiment precede changes in numerical ratings?

- H_0 (null): There is no monotonic association between average sentiment in month t and average rating in month t+1.
- H_1 (alternative): Higher average sentiment in month t is associated with higher average ratings in month t+1.

Sentiment extracted from review text can reflect changes in visitor experience earlier than star ratings. By testing whether sentiment in one period is associated with ratings in the following period, we assess its potential as an early-warning signal. If sentiment shifts precede rating changes, destination managers can identify emerging issues and intervene before declines become visible in numerical ratings.

A Spearman rank correlation reveals a strong monotonic relationship between average sentiment in month t and average rating in month t+1, indicating that shifts in expressed sentiment are closely followed by changes in numerical evaluations.

The null hypothesis is rejected at conventional significance levels.
The findings show a strong and statistically significant positive association between sentiment and subsequent ratings. Periods of more positive sentiment are consistently followed by higher average ratings in the next month, while sentiment deterioration precedes rating declines.

```
H1 Spearman correlation: 0.735
H1 p-value: 1.3606e-08
```

Figure 9. Output of the Spearman rank correlation

H2 – Impact of Crowding on Satisfaction: Do attractions with frequent crowding complaints receive lower ratings during peak season?

- H_0 (null): There is no difference in visitor ratings between reviews mentioning crowding and reviews not mentioning crowding during peak season.
- H_1 (alternative): Reviews mentioning crowding during peak season are associated with lower visitor ratings than reviews without crowding complaints.

Crowding is a recurrent experiential issue in tourist attractions, particularly during peak periods. By comparing ratings from reviews that mention crowding with those that do not, this hypothesis assesses whether overcrowding has a measurable negative effect on visitor satisfaction. If crowding complaints are linked to lower ratings, this would indicate that congestion is not only frequently mentioned but also materially affects perceived quality.

A Mann–Whitney U test reveals a clear difference between the two groups. Reviews that mention crowding during peak season exhibit systematically lower rating distributions than reviews without crowding mentions. The null hypothesis is rejected at conventional significance levels.

The findings indicate a strong and statistically significant negative association between crowding complaints and visitor ratings during peak season. Reviews that reference crowding consistently report lower satisfaction levels, confirming that overcrowding materially harms the visitor experience.

```
H2 U-statistic: 9168164
H2 p-value: 7.8345e-84
```

Figure 9. Output of the Mann–Whitney U test

From an evaluation perspective, this result validates crowding mentions as a meaningful experiential signal rather than incidental commentary. Managing congestion during peak periods emerges as a critical operational lever for protecting visitor satisfaction and preventing rating deterioration.

6. DISCUSSION

This project aimed to support Turismo de Lisboa by converting large-scale Tripadvisor reviews into actionable insights for experience management and marketing strategy. By integrating text mining, behavioral analysis, similarity measures, and hypothesis testing, the analysis moves beyond ratings to explain what drives visitor satisfaction and dissatisfaction.

Addressing the Key Business Questions

The results confirm that numerical ratings alone provide an incomplete view of visitor experience. Ratings are highly skewed and slow to react to emerging issues, whereas text-based indicators such as sentiment and topic mentions reveal early signals of dissatisfaction. Operational issues—particularly crowding and service-related problems—are consistently associated with lower satisfaction, making them priority areas for intervention.

Strategic Insights from Modelling

The modelling approaches reinforce the value of differentiated strategies. Sentiment and topic analysis uncover specific experience drivers that can be acted upon operationally. RFM analysis indicates that a limited number of highly engaged reviewers influence reputation more strongly than the rest, supporting a targeted engagement strategy. Similarity analysis highlights opportunities for cross-promotion and demand redistribution by recommending lesser-known attractions with similar experience profiles to reviewers.

Finally, market basket analysis reveals which attractions are frequently co-mentioned within the same trips, informing bundling strategies, itinerary design, and joint marketing initiatives that reflect how visitors actually experience the city.

Hypothesis-Driven Evaluation
The hypothesis testing phase provides statistical support for the exploratory and modelling findings. Sentiment is strongly and significantly aligned with ratings, validating its use as an early-warning signal. Additionally, reviews mentioning crowding during peak periods are associated with significantly lower ratings, confirming crowding as a key determinant of visitor satisfaction rather than a neutral contextual factor.

Alignment with Project Objectives
Overall, the results align closely with the project objectives: improving experience quality monitoring, identifying operational levers with the highest impact, supporting segmentation, and enabling more proactive destination management. The findings confirm that review-based analytics are a valuable input for strategic and operational decision-making in sustainable tourism.

Insights to Action

1 - Top 5 urgent fixes

- **Crowding Alerts (Operations Team):** Install signage or digital updates to manage flow during Peak Season. **Rationale:** Validated H2 confirmed that crowding complaints in peak months cause significant rating drops.

- **Rapid Response Protocol (Reputation Manager):** Mandate <48h reply for all review, regardless rating. **Rationale:** Current response times (approx. 20 days) are too slow, risking reputational damage.

- **Queue Training (HR/Website Manager):** Train staff or create software specifically on managing entry lines and waiting visitors. **Rationale:** Association rules show "Staff" and "Wait/Queue" frequently co-occur in negative reviews.

- **Reply Coverage Audit (Digital Marketing):** Ensure top 20 sites have active management accounts. **Rationale:** Market Basket analysis shows replies drive engagement, yet only ~34% of reviews currently receive one.

- **Value-for-Money Audit (Quality Assurance):** Review pricing and product for attractions rated ≤3 stars. **Rationale:** Text mining identifies "Price" as a primary keyword in negative reviews, signaling a value gap.

2 - Medium-Term Investments

- **RFM Retention Programme:** Launch targeted campaigns for the "At Risk" segment (recently active but declining). **Rationale:** RFM analysis identified this segment as the highest short-term return opportunity to prevent churn.

- **"Hidden Gems" Bundling Strategy:** Create digital bundles pairing top attractions with similar, less-visited ones. **Rationale:** Similarity Analysis successfully identified high-quality peers for popular sites. This addresses the risk that 24% of all reviews concentrate on a single attraction.

- **Staff Excellence Programme:** Launch soft-skills training for tour guides and on-site staff. **Rationale:** The Network Graph identifies "Staff" as a central hub connecting to all rating levels. It is the single biggest operational lever for satisfaction vs. dissatisfaction.

3 - Marketing Messaging

- **"The People of Lisbon" (Human Connection):** Highlight specific guides/staff in campaigns. **Why:** Text Mining confirms that while queues drive negativity, "Staff" is the primary driver of positive sentiment and 5-star ratings

- **"Smart Travel" (Seasonality):** Promote April/October (Shoulder Season) visits. **Why:** Seasonality Analysis shows volume peaks in summer, causing the crowding pressure that lowers satisfaction. Shifting demand protects the experience quality.

- **"Re-Discover Lisbon" (Loyalty):** Target the "Recent but Low-Engagement" segment. **Why:** RFM Analysis identified this as the largest active reviewer segment (8.97%), converting them to high-value advocates is the highest ROI marketing move.

4 - Monitoring: Governance & Experience Quality Index

- **Monthly Governance Meeting:** Review the city-level Experience Quality Index (EQI).

Dashboard Metrics:

- **1 - Sentiment Early Warning (H1):** Monitor the Sentiment Score trend. Action: If sentiment drops, intervene immediately. **Rationale:** H1 confirmed that sentiment drops precede rating drops.

- **2 - Crowding Pressure:** Track % of reviews mentioning "Queue/Crowd". **Threshold:** If >10% of weekly reviews, trigger capacity protocols.

- **3 - Response Velocity:** Target >80% of negative reviews answered within 7 days (improving current ~20-day average).

Limitations

Despite the richness of the dataset, certain data limitations affected the extent to which all project objectives could be addressed as originally intended. These constraints primarily relate to missing information and limitations in temporal granularity, which influenced the type and depth of analyses that could be reliably conducted. The main limitations we had are specifically listed below.

Overall
The project's description and objectives don't add up to the data we were provided with, mainly.

Location
In order to derive meaningful insights using geographic location, we would need to have enough location/country values. This was not the case, as the dataset had around 87% and 60% of missing values in these variables, respectively.

Time/Date
We don't have data regarding the day of the visit, only the month and year. On the other hand, we have the day, month and year of the review. We observed that in almost every case, the review was made days or even weeks after the visit, so imputing the day of the review in the visit date isn't a viable option. In fact, the variable 'reviewDate' was only useful to extract the month into the visit_date missing values, as most reviews were made within the month of the visit; and to derive the 'fast_reply' variable. Therefore, in a later stage of our analysis, the variable 'reviewDate' was no longer of any use to us, urging us to delete it.